Find Original Array From Doubled Array (medium)

Problem Statement

A doubled array `changed` is formed from the `original` array after appending twice the value of each element in the `original` array and `randomly shuffling` elements in the updated `original` array.

Given an array `changed`, return the `original` array if `changed` is a valid `doubled` array. Otherwise, return `empty` array.

Examples

- **Example 1**:
  - **Input**: `changed = [2,4,1,8]`
  - **Expected Output**: `[1,4]`
  - **Justification**: The original array [1,4] is doubled to form [2,4,1,8] (1's double is 2, and 4's double is 8).
- **Example 2**:
  - **Input**: `changed = [6,3,0,0]`
  - **Expected Output**: `[0,3]`
  - **Justification**: The original array [0,3] is doubled to form [6,3,0,0] (0's double is 0, and 3's double is 6).
- **Example 3**:
  - **Input**: `changed = [1, 2, 2, 4, 4, 8, 16, 32]`
  - **Expected Output**: `[1, 2, 4, 16]`
  - **Justification**: The original array [1, 2, 4, 16] is doubled to form [1, 2, 2, 4, 4, 8, 16, 32] (1's double is 2, 2's double is 4, 4's double is 8, and 16's double is 32).

Try it yourself

Try solving this question here:

